

10/09/26

Signal Processing

o Recap: $x^*[n] \leftrightarrow x^*(e^{-j\omega})$

o $e^{j\omega n}$ and LTI systems

o DFT properties: $x^*[-n]$, $x_e[n]$, $x_o[n]$, $\text{Re}\{x[n]\}$, $j\text{Im}\{x[n]\}$, real signals

o Let $h[n]$ be an LTI system's impulse response. Find the response of this

system to $\underbrace{x[n]} = \underbrace{e^{j\omega n}}$.

- $y[n] = x[n] * h[n]$

$$= \sum_{k=-\infty}^{\infty} x[k] \cdot h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} e^{j\omega k} \cdot h[n-k] = \sum_{k=-\infty}^{\infty} e^{j\omega k + \underbrace{\omega n - \omega n}} h[n-k]$$

$$y[n] = e^{j\omega n} \cdot \underbrace{H(e^{j\omega})} ; \text{ i.e.; the output is a scaled version of the input where the scaling factor is } H(e^{j\omega})$$

$\Rightarrow e^{j\omega n}$ is an eigen function of LTI systems.